

UPSC

Name :

Q.1 Do not
Write in
this
margin

Artificial Intelligence (AI) and Precision Farming have been proposed as solutions to improve agricultural productivity in India. Critically analyze their feasibility and challenges in the Indian agricultural landscape.

Do not
Write in
this
margin

India's Agricultural productivity is 30% lower than the Global average due to various factors such as small land holdings, traditional farm practices and soil degradation etc.

And to overcome this situation AI driven Precision Farming being seen as beacon of hope.

Role of AI and Precision Farming

Pre harvest time

Seed Biotechnology

- ↳ Development of High yielding Seed, Climate resilient and pest resistant Seed by CRISPR with use of AI.

Feasibility & Challenges:

- ↳ Availability of these high valued Seeds may not be feasible for small and marginalised farmers.
- ↳ Awareness and digital illiteracy may become challenges.

Name :

Do not
Write in
this
margin

Soil health Monitoring:

↳ A.I driven soil health monitoring will reduce soil degradation and promote optimal soil utilization.

feasibility and challenges:

↳ Infrastructural facility of soil monitoring standards at remote location undermine its feasibility.

↳ Departure from traditional farm practices and mono crop culture will may create challenges, Specifically in constrained resources.

Pesticide & Fertilizers:

↳ Drone based pesticide application will ensure optimal requirement of pesticide & fertilizers.

feasibility & Challenges:

↳ Easy ^{access} and cheap availability of drones for marginal farmers.

↳ Regulatory measures to monitor drone use at mass scale may face challenges.

Do not
Write in
this
margin

Name :

Do not
Write in
this
margin

Irrigation

Do not
Write in
this
margin

- ↳ Departure from flood irrigation.
- ↳ field level water management and smart micro-irrigation system will lead to optimal and sustainable use of water resources
- ↳ As per book "Water for food & water for life" it has been said that Country applying smart irrigation system have 50% water conservation and 25-30% in yield growth.

feasibility & Challenge

- Cost of modern A.L driven drip & water sprinkler machines.
- water unavailability in drought area.

Weather monitoring

- AI based precision weather monitoring will enhance crop productivity by minimizing ^{loss due to} ~~to~~ erratic climate pattern.

feasibility & Challenges:-

- Unavailability of remote weather management system.
- Lack of its outreach to small & marginal farmers.

Name :

Do not
Write in
this
margin

Post Harvest time

Do not
Write in
this
margin

Modern Harvesting Technology

- Use of state of the art, sensor based AI driven harvesting machinery will reduce harvest losses and also ensure farm waste management.

Feasibility & Challenges:

- These technology and advanced machinery will not be feasible to fragmented land holdings.
- Impact of these technology will rise import bills.

Food processing

- As per FAO-2023 report India loses about 30000 Cr \$ in post harvest.
- AI based precision food processing and value addition will minimize losses and increase productivity.
- AI driven Cold Chain system for perishable crop will increase shelf life and reduce loss.

Feasibility and Challenges

- It may be feasible to apply value addition and will also be export.
- Lack of infra food processing infra in remote location will be challenges.

Name :

Do not
Write in
this
margin

Market Linkage

Do not
Write in
this
margin

- Proper market linkage mechanism will ensure integration in Global Value Chain and enhance farmer income.
- AI driven demand and supply forecasting will provide equilibrium time to sell harvest at optimum level.

Feasibility & Challenges:

- Rural-urban digital divide will may retard this implementation.
- Big Corporates and Intermediaries will may gain all of this market.

Fin tech Solution:

- As per the book "The Great Convergence", if rural ecosystem connected with digital connectivity and multilateral financing then rural migration will reduce and its economy will increase.

Challenge

↳ Financial inclusion at grassroots level is still a challenge.

↳ Credit accessibility and infrastructural support

Initiatives by Govt to Overcome Challenges

↳ PM Shan-Shree Krishi Yojana - 100 low production district selected for this.

↳ Establishment of Mahana Board in Bihar

UPSC

Name :

Do not
Write in
this
margin

After analysing the role and effectiveness of A.I. driven precision farming it may become Game changer for Agri-productivity at mass level. ~~the government~~ ~~schemes and policies towards this~~
But to achieve the transformative potential of all the modern initiatives will achieve its goal if ~~bureaucracy~~ bottlenecks break down.

Do not
Write in
this
margin

Name :

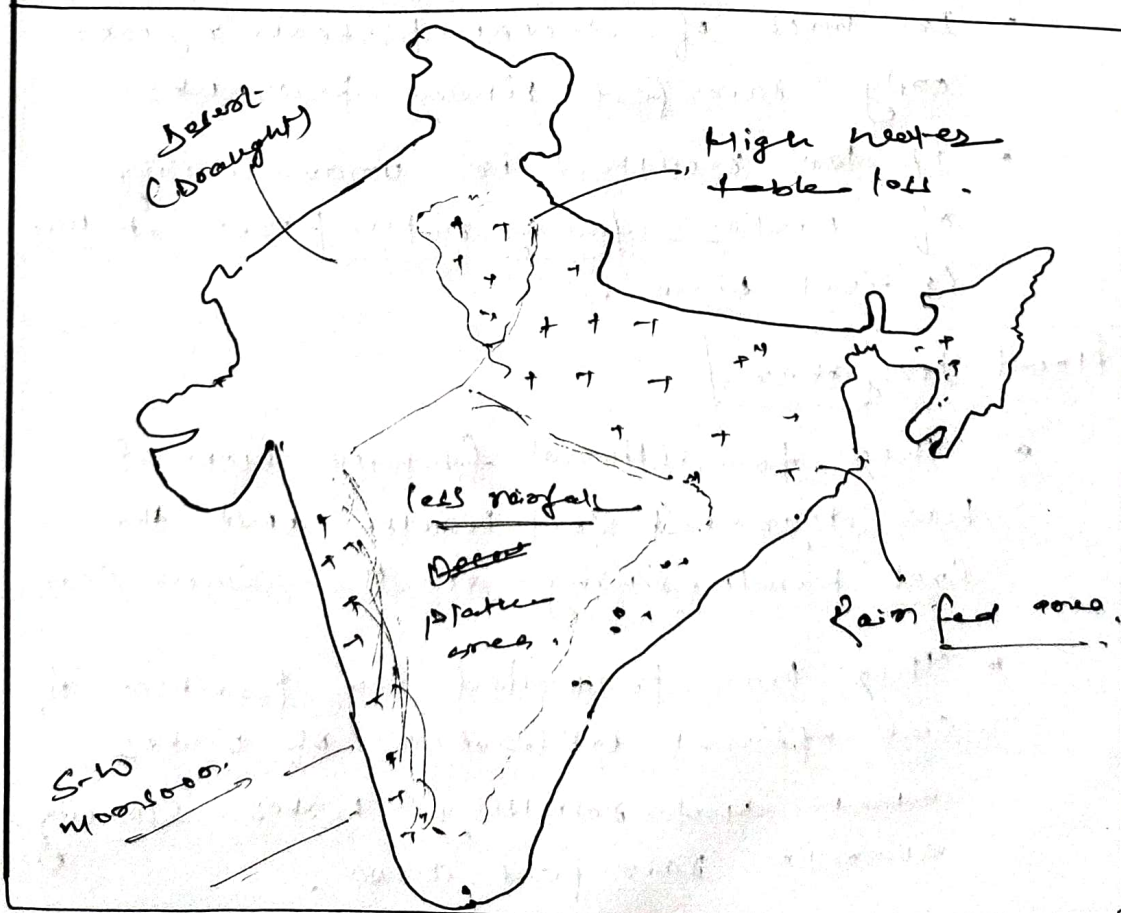
2.

Do not
Write in
this
margin

Water Scarcity is one of the major Challenges in Indian agriculture. Evaluate the role of micro-irrigation techniques in addressing this issue and discuss the policy measures needed for their widespread adoption.

Do not
Write in
this
margin

Indian economy is rain fed economy. 75% of its share depends upon South-West monsoon. But ~~trad~~ fixed irrigation prevalent in traditional farm practices has led to loss of water table and less productivity. Micro-irrigation ~~may~~ will provide a technique for optimal utilization and sustainable use of water.



Name :

Do not
Write in
this
margin

Do not
Write in
this
margin

Water Scarcity as a Challenge:-

Erratic Climate Pattern

- As per recent IMO report due to erratic change in climatic pattern of EL-NINO & LA-NINA, rainfall has drastically changed.
- Absence of reservoir for forest water restoration not done properly.
- Erratic pattern leads to Flash floods and droughts frequently degrading the agro-productivity.

Non perennial drainage system

- In most of deccan plateau region only rain fed rivers present.
- It ~~has~~ resulted in unavailability of water for agri-purposes at this critical time.

Flood Irrigation

- This traditional farming method has degraded soil health and the crop productivity at the large scale.
- This has prevailed the practice of sub-optimal utilization of water ~~even~~ and resulting water scarcity even in rain fed area.

Name :

Do not
Write in
this
margin

Tube well irrigation.

- Beyond rain fed area where drainage system unavailable, this zone is predominantly
- Most critical point is that its flood irrigation technique has resulted in depletion of water table in already water scarce area.

Do not
Write in
this
margin

Role of Micro-irrigation technique :-

Optimal and Sustainable Utilization.

- Dedicated micro-irrigation water technique will promote optimal & sustainable use of water in water scarce region.

Higher Crop yield.

- It will lead in higher crop yield and prevention of seed wash away losses & fertilizer losses.

Climate resilience

- Micro-irrigation technique uses directly less amount of water hence it will ensure ^{resistance against} ~~rain~~ rainfall shocks and climate resilience.

Smart Irrigation

- field level water management will lead to smart irrigation in modern era,

Name :

Do not
Write in
this
margin

- As per book "Water for food & water for life", Smart irrigation system applied Countries has ensure its 50% water conservation and 25-30% yield growth.

Do not
Write in
this
margin

Policy measures needed for widespread adoption :-

1. Promote awareness about Smart irrigation system, because as per NABARD-2022 report more than 60% of irrigated land still using traditional flood irrigation which leading to water waste and low productivity.
2. Provision of tax incentives to drift irrigation sprinkler machinery companies.
3. Provision of schemes that ensure subsidies to marginalised farmers for input cost support.
4. Implement micro-irrigation technique to most drought effected areas as a pilot project then spread its outreach.
5. Policy to develop small scale rain-water harvesting at mass scale for water conservation.

UPSC

Page No. :

Roll No. :

Name :

Do not
Write in
this
margin

At last if synchronised ^{effort}~~effect~~ of govt, prvt sector and small farmer proceed then micro-irrigation will ~~be~~ paved the way ~~of~~ for solution of Indian agricultural productivity. And it's not only about productivity, a sustainable use of water ~~with~~ only will water and food security for next generation.

Do not
Write in
this
margin